

NAFISA BERLIANA INDAH PRATIWI

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EDUCATION

IPB University | GPA: 3.42

Jul 2021 - Aug 2024

M.Si. | *Statistics and Data Science*

Bogor, Indonesia

Universitas Sebelas Maret | GPA: 3.16

Aug 2016 - April 2020

S.Mat. | *Mathematics*

Surakarta, Indonesia

DQLab

June 2024 - October 2024

Bootcamp | *Data Analyst with Python and SQL*

Bootcamp | *Machine Learning & AI for Beginner*

International Language Program (ILP)

May 2025

TOEFL ITP | *Score: 553*

EXPERIENCE

Freelance Quantitative Analyst

2021 - Present

Remote

- Conducted quantitative analysis with a focus on econometric modeling, complemented by statistical inference, exploratory data analysis, and machine learning techniques across diverse social research topics and policy briefs.
- Created visualizations and dashboards using ggplot2 and Google Looker Studio to support data-driven insights.
- Designed conceptual diagrams and flowcharts to map analytical workflows and research logic; proficient in Microsoft Visio and related tools.
- Contributed to academic writing, including structuring papers, refining content, and improving language clarity for publication or thesis submission.
- Designed survey instruments, selected appropriate methods, and performed end-to-end data analysis with emphasis on research validity.

Instructor Assistant

Mar 2022 - Present

Quantitative Research Lab, Department of Economics, IPB University

- Assisted participants during hands-on sessions on applied econometrics using RStudio, organized by the Quantitative Research Lab, Department of Economics.
- Guided participants in implementing regression models, time series analysis, and diagnostic tests using R packages.
- Clarified statistical concepts and helped troubleshoot common coding issues in R during guided exercises.

Book Author

Dec 2023

- Contributed to the book *Metode Kuantitatif dengan RStudio: Penerapan untuk Riset Manajemen dan Bisnis*.
- Authored the chapter on cluster analysis and contributed to additional sections on applied R-based methods.

Teaching Assistant – Data Exploration

Jul 2020 – Dec 2020

Universitas Sebelas Maret (UNS), Indonesia

- Provided academic support in Exploratory Data Analysis, guiding students through coursework and interpretation.
- Assisted final-year students in the Soft Computing Research Group with thesis development and progress tracking.

Intern – Curriculum & Talent Analysis

Jan – Jul 2019

Al-Firdaus Islamic School / PT. Tiga Serangkai Pustaka Mandiri

- Evaluated student interests and talents through interviews and academic records within the International Baccalaureate framework.
- Recommended curriculum adjustments to enhance alignment with student profiles and learning outcomes.

PROJECTS

City Branding Research

Jun 2025 - Present

Independent Project

- Designing a comprehensive city branding questionnaire covering key indicators such as cultural identity, economic competitiveness, stakeholder engagement, and digital presence.
- Developing Likert-scale survey instruments targeted at multiple stakeholder groups including government, academics, business owners, and cultural figures.
- Constructing an analytical framework grounded in city branding theory (e.g., place identity, heritage value, economic positioning).
- Currently collecting and analyzing stakeholder perception data alongside secondary sources, including Statistics Indonesia (BPS) and regional planning documents.
- Visualizing preliminary findings and documenting research progress using Quarto for iterative feedback and future policy application.

Smart City Index Research

Feb 2025 - June 2025

Independent Project

- Designed a tailored survey instrument to assess smart city dimensions, incorporating national and theoretical constructs.
- Implemented Analytic Hierarchy Process (AHP) with expert respondents to derive weighted scores across key pillars.
- Developed an original evaluation framework by synthesizing Kemenkominfo's national indicators with the Giffinger model.
- Collected and visualized secondary data from regional governments and Statistics Indonesia (BPS) to contextualize dimension-level scores.
- Produced cross-framework insights revealing structural alignment and divergence between national policy and global smart city theory.

Microplastic Perception Study

Dec 2024

Quantitative Analyst (Contributor)

Makassar, Indonesia

- Conducted behavioral clustering analysis using k-prototype algorithm on household survey data related to microplastic awareness and actions.
- Applied Silhouette Index for optimal cluster selection, identifying two distinct behavioral-geographic segments within the urban district.
- Implemented Random Forest-based feature importance analysis to interpret key drivers such as education level, environmental attitudes, and length of residence.
- Interpreted results in relation to geographic features (e.g., landfills, lakes) and identified socio-environmental disparities in engagement with plastic pollution.
- Contributed to research recommendations for geographically targeted public awareness campaigns and policy interventions.

UEQ-Based UX Analysis - User Experience Evaluation

Sept 2024 - Nov 2024

Independent Academic Project

- Supported a UX research project by analyzing website evaluation data using the standardized User Experience Questionnaire (UEQ), which measures six core dimensions: attractiveness, perspicuity, efficiency, dependability, stimulation, and novelty.
- Applied non-parametric hypothesis testing (Kruskal-Wallis test) to identify significant differences in user experience ratings across multiple websites.
- Interpreted patterns in ordinal-scale responses and summarized key findings to assist in academic reporting and usability recommendations.

AWARDS & PRESENTATIONS

Winner – Call for Papers

2023

Sidang Pleno ISEI XXIII & Seminar Nasional

- Paper Title: Interkoneksi Pertumbuhan Ekonomi Antar Provinsi di Indonesia

- Selected as one of the best papers through national-level peer review and presented during the plenary session of the annual ISEI conference.

Presenter – International Academic Paper Presentation <i>Sidang Pleno ISEI XXIII & Seminar Nasional</i>	2023
Presenter – International Academic Paper Presentation <i>International Conference on Mathematics and Learning Research (ICOMER)</i>	2021
Presenter – International Academic Paper Presentation <i>International Conference of Mathematics and Mathematics Education (ICMME)</i>	2021
Presenter – International Academic Paper Presentation <i>International Conference on Mathematics: Education, Theory, and Application (ICMETA)</i>	2020
Presenter – International Academic Paper Presentation <i>Ahmad Dahlan International Conference on Mathematics and Mathematics Education (ADINTERCOMME)</i>	2019

PUBLICATIONS

Nafisa Berliana Indah Pratiwi, Indahwati, F. Anwar, “[Village Potential Mapping: Comprehensive Cluster Analysis of Continuous and Categorical Variables with Missing Values and Outliers Dataset in Bogor, West Java, Indonesia](#)”, *Scientific Journal of Informatics*, 2024.

Nafisa Berliana Indah Pratiwi, D.R.S. Saputro, P. Widyaningsih, “[Fuzzy c-shells for clustering of data’s LQ45 in Indonesia based on earnings per share and price earning ratio](#)”, *Journal of Physics: Conference Series*, 2021.

Nafisa Berliana Indah Pratiwi, D.R.S. Saputro, “[Fuzzy c-shells clustering algorithm](#)”, *Journal of Physics: Conference Series*, 2020.

D.R.S. Saputro, P.H. Utomo, H. Prasetyo, **Nafisa Berliana Indah Pratiwi**, “[Evaluation of hybrid algorithms \(constraint base and bound and collapse\) in building Bayesian networks structures](#)”, *AIP Conference Proceedings*, 2022.

D.R.S. Saputro, N. Aulia, P.Y. Astuti, **Nafisa Berliana Indah Pratiwi**, S. Sutanto, “[Error correction models \(Case of data inflation in Indonesia\)](#)”, *AIP Conference Proceedings*, 2022.

D.R.S. Saputro, **Nafisa Berliana Indah Pratiwi**, R. Kusumawati, “[Logistic smooth transition autoregressive model parameter estimation using Gauss Newton](#)”, *AIP Conference Proceedings*, 2022.

D.R.S. Saputro, A. Susanti, **Nafisa Berliana Indah Pratiwi**, “[The handling of overdispersion on Poisson regression model with the generalized Poisson regression model](#)”, *AIP Conference Proceedings*, 2021.

D.R.S. Saputro, Y.K. Wardani, **Nafisa Berliana Indah Pratiwi**, P. Widyaningsih, “[Data simulation with Markov Chain Monte Carlo, Gibbs sampling, and Bayes \(beta-binomial\) methods as the parameter estimations of spatial bivariate probit regression model](#)”, *AIP Conference Proceedings*, 2021.

I.L. Shabrina, D.R.S. Saputro, **Nafisa Berliana Indah Pratiwi**, P. Pangadi, “[Liquid 45 modeling based on Markov chain and fuzzy linguistic summary using fuzzy stochastic](#)”, *AIP Conference Proceedings*, 2021.

SKILLS

Expertise: Econometrics, Statistical Inference, Exploratory Data Analysis, Survey Design, Data Visualization, Machine Learning, Clustering Analysis (e.g. k-means, k-modes, k-prototype, hierarchical, soft clustering), Time Series Analysis, Hypothesis Testing, Policy Analysis

Software: R (advanced), SPSS, Stata, Microsoft Office, Microsoft Visio, Google Looker Studio, SQL (basic), Quarto, LaTeX, Git

Languages: R, Python (basic), SQL (basic)